



Factors Associated with Very Late Presentation Among PLHIV in Southern Jamaica



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Introduction

Despite programmatic efforts, advanced HIV continues to be a challenge within the Caribbean region which increases the risk of mortality, and onward transmission.

Methods

A retrospective cohort study was carried out on persons living with HIV linked to health facilities in Southern Jamaica during 2017 to 2022 to explore factors associated with stage of HIV diagnosis. Patients were classified into four categories based on the WHO guidelines for staging of CD4. Descriptive statistics and chi-square tests analyzed factors associated with CD4 stage.

Results

Of the 410 patients, 35% were “very early” (CD4 $\geq$ 500 cells/ μ l), 20% were “fair early” (CD4 350-499 cells/ μ l), 23% were “late” (CD4 200 - 349 cells/ μ l) and 22% were “very late” (CD4 < 200 cells/ μ l). The study revealed that age was associated with stages of CD4 ($p < 0.01$). 7% aged 15-24yrs were diagnosed “very late”, compared to 23% aged 25-34yrs, 26% aged 35-49yrs and 28% aged 50 and over. Gender was also associated with stage of CD4 ($p < 0.01$). Males were more likely to be diagnosed very late (32%) compared to females (15%). Persons considered “alive” 20% were diagnosed “very late”, compared to 33% who migrated and 50 % considered deceased.

Results Cont'd

Table 1- Sociodemographic factors and Treatment outcomes of PLHIV linked to care in the Southern Region of Jamaica between 2017-2022 stratified by the categories of CD4 staging Early presenters- CD4 $\geq$ 500 cells/ μ l (Stage 1), Fairly Early 350-499 cells/ μ l (Stage 2), late presenters 200 - 349 cells/ μ l Stage 3) & Very late presenters are < 200 cells/ μ l (Stage 4).

Sociodemographic Variable			Categories of CD4 Staging									
	Total (n)	% of Toal	Very Late: Stage 4 (<200)		Late: Stage 3 (200-349)		Fairly Early: Stage 2 (349-499)		Early: Stage 1 (≥500)		Chi-square	P-value
	410		n=92 or 22%		n=93 or 23%		n=81 or 20%		n=144 or 35%			
Gender	410		n	%	n	%	n	%	n	%	25.641	<0.01
Female	230	56.1%	35	15%	48	21%	45	20%	102	44%		
Male	180	43.9%	57	32%	45	25%	36	20%	42	23%		
Age Group											41.907	<0.01
15-24	70	17.1%	5	7%	8	11%	18	26%	39	56%		
25-34	92	22.4%	21	23%	17	18%	13	14%	41	45%		
35-49	138	33.7%	36	26%	35	25%	35	25%	32	23%		
50+	109	26.6%	30	28%	33	30%	15	14%	31	28%		
Parish	409										8.631	0.195
St. Elizabeth	92	22.4%	25	27%	19	21%	19	21%	29	32%		
Manchester	136	33.2%	37	27%	28	21%	21	15%	50	37%		
Clarendon	182	44.4%	30	16%	46	25%	41	23%	65	36%		
Year of Linkage	410										20.388	0.158
2017	61	14.9%	13	21%	10	16%	12	20%	26	43%		
2018	83	20.2%	21	25%	25	30%	11	13%	26	31%		
2019	40	9.8%	15	38%	7	18%	5	13%	13	33%		
2020	72	17.6%	15	21%	13	18%	17	24%	27	38%		
2021	70	17.1%	14	20%	12	17%	18	26%	26	37%		
2022	84	20.5%	14	17%	26	31%	18	21%	26	31%		
Year of Diagnosis	410										20.825	0.143
2017	68	16.6%	14	21%	12	18%	12	18%	30	44%		
2018	78	19.0%	21	27%	24	31%	11	14%	22	28%		
2019	42	10.2%	15	36%	7	17%	6	14%	14	33%		
2020	71	17.3%	15	21%	12	17%	17	24%	27	38%		
2021	67	16.3%	14	21%	12	18%	17	25%	25	37%		
2022	84	20.5%	14	17%	26	31%	18	21%	26	31%		
Outcomes	410										22.139	<0.01
Dead	28	7%	14	50%	10	36%	2	7%	2	7%		
Migrated	6	1%	2	33%	1	17%	1	17%	1	17%		
Alive	376	92%	76	20%	82	22%	78	21%	140	37%		
	Total (n)		Very Late: Stage 4 (<200)		Late: Stage 3 (200-349)		Fairly Early: Stage 2 (349-499)		Early: Stage 1 (≥500)		Chi-square	P-value
Retention In Care (Alive)	376		n	%	n	%	n	%	n	%	3.104	0.376
Retained in Care	315	84%	52	17%	70	22%	64	20%	113	36%		
Lost to Follow Up	61	16%	8	13%	12	20%	14	23%	27	44%		
Treatment & Suppression Status	273										2.297	0.513
Alive + ART Suppressed	244	89%	48	20%	61	25%	48	20%	87	36%		
Alive + ART Unsuppressed	29	11%	9	31%	6	21%	6	21%	8	28%		

Conclusions

Factors such as gender, age at diagnosis, and patients now deceased were associated with very late presentation. Strengthening HIV testing targeting men and conversations on sexual health with older adults during clinic visits can reduce very late presentation.

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